

Technical Requirement Document

Here is the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

TR-DLT-001: Load Customer and Order Data into Delta Tables

Related Functional Requirement(s): FR-DLT-001

Objective: Implement a Delta Live Table to load customer and order data from CSV files into Delta tables.

Target Cluster Configuration:

- Cluster Name: Gen AI POC Databricks Cluster
- Databricks Runtime Version: 10.4 LTS
- Node Type: Standard_DS3_v2
- Driver Node: Standard_DS3_v2
- Worker Nodes: 2-4 Standard_DS3_v2
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: delta-lake, spark-csv

Source Data Details:

- Dataset Name: Customer Data
- Location: ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata``
- Format: CSV
- Description: Customer data in CSV format
- Dataset Name: Order Data
- Location: ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata``
- Format: CSV
- Description: Order data in CSV format

Target Data Details:

- Output Name: Customer Delta Table
- Location: ``gen_ai_poc_databrickscoe.sdlc_wizard.customer``
- Format: Delta
- Description: Customer data in Delta format
- Output Name: Order Delta Table

- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.order`
- Format: Delta
- Description: Order data in Delta format

Job Flow / Pipeline Stages:

- Read customer CSV data and load into Customer Delta Table
- Read order CSV data and load into Order Delta Table
- Add TotalAmount column to Order Delta Table

Data Transformations / Business Logic:

- Step: Load customer data
- Description: Read customer CSV data and load into Customer Delta Table
- Transformation Logic: Use Spark CSV reader to read CSV data and write to Delta table
- Step: Load order data
- Description: Read order CSV data and load into Order Delta Table
- Transformation Logic: Use Spark CSV reader to read CSV data and write to Delta table
- Step: Calculate TotalAmount
- Description: Add TotalAmount column to Order Delta Table
- Transformation Logic: Use Spark to calculate TotalAmount as product of PricePerUnit and Qty

Error Handling and Logging:

- Log errors to a Delta table for auditing and debugging purposes
- Use try-except blocks to handle exceptions during data loading and transformation

TR-DLT-002: Clean and Transform Customer and Order Data

Related Functional Requirement(s): FR-DLT-002

Objective: Implement data quality checks to remove null and duplicate records from Customer and Order Delta tables.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details: Customer Delta Table, Order Delta Table

Target Data Details: Cleaned Customer Delta Table, Cleaned Order Delta Table

Job Flow / Pipeline Stages:

- Remove null records from Customer and Order Delta tables
- Remove duplicate records from Customer and Order Delta tables

Data Transformations / Business Logic:

- Step: Remove null records
- Description: Remove null records from Customer and Order Delta tables

- Transformation Logic: Use Spark to filter out null records
- Step: Remove duplicate records
- Description: Remove duplicate records from Customer and Order Delta tables
- Transformation Logic: Use Spark to drop duplicates based on primary keys

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-003: Create and Load Order Summary Table

Related Functional Requirement(s): FR-DLT-003

Objective: Implement a Delta Live Table to create and load Order Summary table with joined Customer and Order data using SCD type 2.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details: Customer Delta Table, Order Delta Table

Target Data Details: Order Summary Delta Table

Job Flow / Pipeline Stages:

- Create Order Summary Delta Table if it does not exist
- Join Customer and Order Delta tables on CustId
- Load joined data into Order Summary Delta Table using SCD type 2

Data Transformations / Business Logic:

- Step: Join Customer and Order data
- Description: Join Customer and Order Delta tables on CustId
- Transformation Logic: Use Spark to join tables on CustId
- Step: Load data into Order Summary table
- Description: Load joined data into Order Summary Delta Table using SCD type 2
- Transformation Logic: Use Spark to implement SCD type 2 logic

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-004: Update Order Summary Table on Customer Changes

Related Functional Requirement(s): FR-DLT-004

Objective: Implement logic to update Order Summary table when there are changes to Customer table.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details: Customer Delta Table, Order Summary Delta Table

Target Data Details: Updated Order Summary Delta Table

Job Flow / Pipeline Stages:

- Identify changes to Customer table
- Update Order Summary table by making old records inactive and new records active

Data Transformations / Business Logic:

- Step: Identify changes to Customer table
- Description: Identify changes to Customer Delta Table
- Transformation Logic: Use Spark to compare current and previous versions of Customer data
- Step: Update Order Summary table
- Description: Update Order Summary table by making old records inactive and new records active
- Transformation Logic: Use Spark to update Order Summary table based on changes to Customer data

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-005: Create and Load Customer Aggregate Spend Table

Related Functional Requirement(s): FR-DLT-005

Objective: Implement a Delta Live Table to create and load Customer Aggregate Spend table with aggregated data from Order Summary table.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details: Order Summary Delta Table

Target Data Details: Customer Aggregate Spend Delta Table

Job Flow / Pipeline Stages:

- Create Customer Aggregate Spend Delta Table if it does not exist
- Aggregate TotalAmount column from Order Summary table by Name and Date
- Load aggregated data into Customer Aggregate Spend Delta Table

Data Transformations / Business Logic:

- Step: Aggregate data
- Description: Aggregate TotalAmount column from Order Summary table by Name and Date
- Transformation Logic: Use Spark to aggregate data using groupBy and sum aggregations
- Step: Load data into Customer Aggregate Spend table
- Description: Load aggregated data into Customer Aggregate Spend Delta Table
- Transformation Logic: Use Spark to write aggregated data to Delta table

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-006: Implement Delta Live Tables

Related Functional Requirement(s): FR-DLT-006

Objective: Implement Delta Live Tables to execute the data pipeline from customer and order data loading to customer aggregate spend data loading.

Target Cluster Configuration: Same as TR-DLT-001

Job Flow / Pipeline Stages:

- Implement Delta Live Tables to execute the data pipeline

Data Transformations / Business Logic:

- Step: Implement Delta Live Tables
- Description: Implement Delta Live Tables to execute the data pipeline
- Transformation Logic: Use Delta Live Tables to orchestrate the data pipeline

Error Handling and Logging: Same as TR-DLT-001